

53. Embryonic connective tissue is made by  
 a) Cartilage      b) Mesenchymal tissue      c) Adipose tissue  
 d) Reticular connective tissue      e) Blood
54. Which connective tissue cell is derived from B lymphocytes and produce antibodies?  
 a) Fibroblast      b) Myofibroblast      c) Histiocyte      d) Plasma cell      e) Mast cell
55. What type of tissue is a tendon composed of?  
 a) Mucous connective tissue      b) Mesenchyme      c) Loose irregular connective tissue  
 d) Dense irregular connective tissue      e) Dense regular connective tissue
56. Which connective tissue cell type is functionally related to the basophiles?  
 a) Fibroblast      b) Myofibroblast      c) Histiocyte      d) Plasma cell      e) Mast cell
57. Which of the following statements is not true about the connective tissue?  
 a) Develop from mesenchyme      b) Has abundant extracellular matrix with dispersed cells  
 c) Has innervation      d) Has no blood supply      e) Is involved in body defense
58. White adipose tissue  
 a) Contains multiple fat droplets      b) Has round centrally located cells  
 c) Is known as multilocular fat cells      d) Is involved in body insulation  
 e) Contains cells that have greater number of mitochondria than that of brown adipose tissue
59. Cells responsible for synthesis of the fibres and ground substance of connective tissue are:  
 a) Macrophages      b) Mast cells      c) Fibroblasts      d) Plasma cells      e) Histiocytes
60. Which of the following cells belong to the mononuclear phagocyt system?  
 a) Macrophages      b) Fibroblasts      c) Mast cells      d) Plasma cells      e) Fibrocyte

## EMBRYOLOGY (40%)

For each of the following Multiple Choice Questions choose the one best appropriate answer and mark "X" in the appropriate box located against the corresponding question number and letter in the separate answer sheet provided. (40 points, 1 point for each correct answer).

1. With regard to spermatogenesis, which statement is false?  
 a. Spermatids are transformed into spermatozoa by spermiogenesis. ✓  
 b. The entire process of spermatogenesis takes approximately 72 hours. X 62-64 days  
 c. Primary spermatocytes are the largest germ cells in the seminiferous tubules.  
 d. Spermatogenesis begins at puberty. ✓      e. Spermatogonia begin to increase in number at puberty. ✓

- Which one of the following statements regarding the exocrine glands is not correct?
- a) Mucous glands occur as tubules, while serous glands occur as acinar ✓
  - b) When mixed mucoserous secretory cells open into a common duct, the mucous cells are found centrally towards the lumen with the serous cells located peripherally. ✓
  - c) Exocrine glands whose secretory end pieces are divided may not be compound glands. ✓
  - d) The goblet cells producing mucus represent unicellular glands. ✓
  - e) Both intercalated and striated ducts are interlobular ducts. ✓

45. Which stage does collagen is released by exocytosis during the process of its synthesis?
- a) Tropocollagen molecules
  - b) Pro- $\alpha$ -chains
  - c) Procollagen molecules
  - d) Collagen fibrils
  - e) Collagen microfibrils

46. In collagen fibrils, striations repeat every ✓ 64-68
- a) 10-15 nm long
  - b) 64-68 nm long
  - c) 100-150 nm long
  - d) 640-680 nm long all other
  - e) 100-150  $\mu$ m long

47. Which type of collagen did you see in the Histology Lab made as bundles in the loose connective tissue? I & III
- a) Type I
  - b) Type II
  - c) Type V
  - d) Type VI
  - e) Type VII

48. Which type of connective tissue makes the stroma of the visceral organs?
- a) White adipose tissue
  - b) Brown adipose tissue
  - c) Reticular connective tissue
  - d) Elastic connective tissue
  - e) Mucoid connective tissue

49. Glycosaminoglycan that is very long and rigid made of thousands of sugar and found in all types of connective tissue is:
- a) Hyaluronic acid
  - b) Chondroitin sulfate
  - c) Dermatan sulfate
  - d) Heparan sulfate
  - e) Keratan sulfate

50. Which of the following is not the nature of ground substance of connective tissue?
- a) Act as mechanical support
  - b) Facilitate rapid diffusion of water-insoluble molecules (water-soluble)
  - c) Facilitate migration of cells
  - d) Act as barrier for micro-organisms ✓
  - e) None of the above

51. Regarding the ground substance of the connective tissue which of the following statements is not correct?
- a) Ground substance acts as a lubricant. ✓
  - b) Glycosaminoglycans attract osmotically active cations -ve  $Na^+$  ✓
  - c) Proteoglycans are made by association of glycosaminoglycans with a protein core. ✓
  - d) Proteoglycan aggregates are formed by association of proteoglycans with glycoproteins. ✓
  - e) Glycoproteins connect one component of extracellular matrix with the other. ✓

52. Regarding the connective tissue fibres:
- a) Elastic fibres make the stroma of many delicate organs. X
  - b) Type IV collagen is the most abundant type of collagen fibres. X (I)
  - c) Type III collagen makes a porous membrane in the basal lamina. (GAG)
  - d) Collagen has high tensile strength with little elasticity.
  - e) Reticular fibers are composed of elastin and fibrillin. (reticular)

elastic fiber = elastin + fibrillin

Proteoglycan = GAG + protein  
except hyaline

Proteoglycan aggregation: proteoglycan + hyaline

35. Which one of the following statement is not true regarding the epithelial tissue?
- a) Cells rest on a basal lamina ✓
  - b) Most of the cells have high turnover. ✓
  - c) Has abundant blood supply. ✓
  - d) Has nerve supply. ✓
  - e) May contain cells that do not belong to the epithelial tissue.

36. Regarding to the structures found on the apical surface of some epithelial cells

- a) Microvilli are supported by a group of microtubules found in their core. ✓
- b) Cilia and flagella are structurally identical, but cilia are longer than flagella. ✓
- c) Sterocilia are related to the microvilli than to the cilia. ✓
- d) The axoneme of cilia is made by a central pair and 9 peripheral triplets of microtubules. ✓
- e) The basal bodies of the cilia appear as terminal bars under the light microscope.

37. Which of the following is located most apically in the junctional complex?

- a) Zonula-adherens
- b) Macula adherens
- c) Zonula occludens ✓
- d) Hemidesmosomes
- e) Gap junction

38. Cell junction that shows dense plaque on the cytoplasmic surface of the cell membrane at the site of junction is which is made by attachment of actin microfilaments and several protein is:

- a) Desmosome
- b) Intermediate junction
- c) Hemidesmosome
- d) Tight junction
- e) Gap junction

39. Cell junction made by a pair of equal disc shaped structural contribution of adjacent cells is: ✓

- a) Desmosome
- b) Intermediate junction
- c) Hemidesmosome
- d) Tight junction
- e) Gap junction

40. How many gap junction proteins known as connexins surround the hydrophilic pore of gap junction? ✓

- a) 3
- b) 6
- c) 9
- d) 12
- e) 15

41. Type of epithelium that covers the luminal surface of distensible organs is:

- a) Transitional epithelium
- b) Simple cuboidal epithelium
- c) Pseudostratified epithelium
- d) Stratified squamous non keratinized epithelium
- e) Stratified squamous keratinized epithelium

42. Glandular cells which release their secretory product along with small apical cytoplasm and plasma membrane are known as:

- a) Holocrine.
- b) Epicrine.
- c) Apocrine.
- d) Merocrine.
- e) Eccrine.

43. Which one of the following is not a correct statement regarding the endocrine glands?

- a) Cells are architecturally arranged as corals, clusters or follicles.
- b) Have more blood capillaries than exocrine glands.
- c) Have a similar manner of initial development with that of exocrine glands as evagination out of the covering epithelium.
- d) Most have their target cells located far away as compared with that of the exocrine glands.
- e) Paracrine cells are secretory cells whose secretory product targets the secretory cells themselves for a feedback mechanism.

11. If a tumor grows into the quadrangular space of the axillary region, which structures would be in danger?

- A. anterior humeral circumflex artery and axillary nerve
- B. posterior humeral circumflex artery and radial nerve
- C. posterior humeral circumflex artery and axillary nerve
- D. radial nerve and profunda brachii artery
- E. profunda brachii artery and axillary nerve

12. If the entire greater tubercle of the humerus was torn away as a result of an injury, which of the following movements of the humerus would be affected?

- a) flexion and abduction
- b) abduction and lateral rotation
- c) flexion and medial rotation
- d) extension and medial rotation
- e) flexion, abduction, and adduction

13. The superficial palmar arterial arch of the hand:

- a) is located in the thenar compartment
- b) is proximal to the deep palmar arterial arch
- c) is deep to the tendons of the flexor digitorum profundus
- d) is formed primarily by the ulnar artery
- e) courses with the deep branch of the ulnar nerve

14. Each of the following muscles connects the upper limb to the thoracic wall, EXCEPT the:

- a) pectoralis major
- b) pectoralis minor
- c) subclavius
- d) latissimus dorsi
- e) serratus anterior

15. Which of the following nerves sends branches to the muscles of the arm?

- A. median
- B. ulnar
- C. dorsal scapular
- D. musculocutaneous
- E. long thoracic

21. Which of the following nerves may be injured in Lower Lesion of Brachial Plexus (Klumpke's Paralysis)?
- suprascapular
  - axillary
  - ulnar
  - dorsal scapular
  - subscapular
22. Which of the following nerves arise directly from the roots of the brachial plexus?
- suprascapular
  - axillary
  - ulnar
  - dorsal scapular
  - subscapular
23. Which of the following arteries arise from the second part of the axillary artery.
- superior thoracic
  - lateral thoracic
  - thoracodorsal
  - subscapular
  - dorsalscapular
24. The thumb action that is **totally** affected by radial nerve trauma is:
- abduction
  - adduction
  - extension
  - flexion
  - opposition
25. The extensor expansion of the fingers is formed by the tendon of the:
- extensor carpi radialis longus
  - extensor pollicis longus
  - extensor digitorum
  - extensor pollicis brevis
  - interossei

3. A high-velocity bullet penetrated the posterior aspect of the shoulder and severely injured the origin of the posterior cord of the brachial plexus. Each of the following nerves might possibly be paralyzed, EXCEPT the:
- a) upper subscapular
  - b) axillary
  - c) thoracodorsal
  - d) radial
  - e) medial pectoral
4. As a result of an injury to the ulnar nerve in the arm, all of the following changes in the hand will be noted, EXCEPT:
- a) claw hand
  - b) the thumb is strongly adducted
  - c) loss of flexion of the distal interphalangeal joints of the fourth and fifth digits.
  - d) There is sensory loss on the little finger and one-half of the ring finger
  - e) There is marked wasting in the hypothenar eminence
5. Sensory loss over the skin covering the dorsal surface of the thumb would indicate damage to which of the following nerves?
- a) deep radial
  - b) superficial radial
  - c) anterior interosseous
  - d) median
  - e) ulnar
6. Compression of the median nerve in the carpal tunnel deep to the flexor retinaculum could affect the functioning of each of the following muscles, EXCEPT the:
- a) second lumbrical
  - b) opponens pollicis
  - c) flexor pollicis brevis
  - d) adductor pollicis

26. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be nonfunctional?

- A. radial
- B. median recurrent
- C. ulnar
- D. deep radial
- E. superficial radial

27. Which nerve is probably damaged if a patient cannot abduct the arm beyond 25 degrees?

- A. axillary
- B. radial
- C. musculocutaneous
- D. median
- E. ulnar

28. Loss of pronation of the hand suggests a lesion of the:

- A. median nerve
- B. radial nerve
- C. ulnar nerve
- D. intercostobrachial nerve
- E. musculocutaneous nerve

29. With an injury to the right upper trunk of the brachial plexus (Erb Duchenne paralysis), the patient's right upper extremity would be:

- A. abducted
- B. elevated
- C. medially rotated
- D. laterally rotated
- E. flexed

30. Loss of opposition of the thumb is a symptom associated with lesion of which nerve?

- A. radial
- B. ulnar
- C. musculocutaneous
- D. median
- E. posterior interosseous